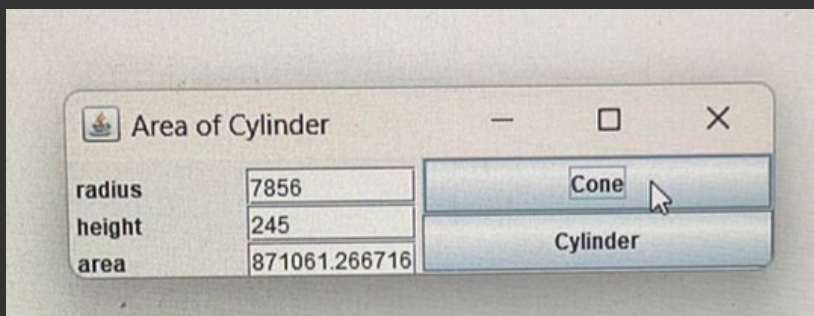




```

practice10_3.java
practice9_quiz.jav
1 import java.awt.*;
2 import javax.swing.*;
3 import java.awt.event.*;
4 public class practice10_3 extends JFrame{
5     double r,h,a;
6     JTextField inp_r = new JTextField(5);
7     JTextField inp_h = new JTextField(5);
8     JTextField otp_a = new JTextField(5);
9
10    public practice10_3(){
11        JLabel r = new JLabel("radius");
12        JLabel h = new JLabel("height");
13        JLabel a = new JLabel("area");
14
15        JPanel p1 = new JPanel(new GridLayout(3,2));
16        p1.add(r);
17        p1.add(inp_r);
18        p1.add(h);
19        p1.add(inp_h);
20        p1.add(a);
21        p1.add(otp_a);
22
23        JButton co = new JButton("Cone");
24        JButton cy = new JButton("Cylinder");
25        JPanel p2 = new JPanel(new GridLayout(2,1));
26        p2.add(co);
27        p2.add(cy);
28
29
30        JPanel mainpanel = new JPanel(new GridLayout(1,2));

```

— GUI

```

31        mainpanel.add(p1);
32        mainpanel.add(p2);
33
34        add(mainpanel);
35
36        co.addActionListener(new co());
37        cy.addActionListener(new cy());
38    }
39    private class co implements ActionListener{
40        public void actionPerformed(ActionEvent e){
41            r = Double.parseDouble(inp_r.getText());
42            h = Double.parseDouble(inp_h.getText());
43
44            a = Math.PI*r*(r+Math.sqrt(h*h+r*r));
45
46            otp_a.setText(Double.toString(a));
47        }
48    }
49    private class cy implements ActionListener{
50        public void actionPerformed(ActionEvent e){
51            r = Double.parseDouble(inp_r.getText());
52            h = Double.parseDouble(inp_h.getText());
53
54            a = (2*Math.PI*r*h)+(2*Math.PI*r*r);
55
56            otp_a.setText(Double.toString(a));
57    }

```

event
ដើម្បីគណនា cone

event
ដើម្បីគណនា cylinder

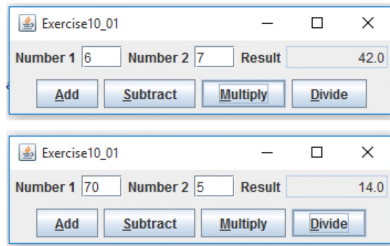
```

58    }
59
60
61
62    public static void main(String[] args){
63        practice10_3 f = new practice10_3();
64        f.setTitle("Exercise 10_03");
65        f.setSize(400,120);
66        f.setLocationRelativeTo(null);
67        f.setVisible(true);
68    }
69 }

```

} ធីតា ហ្វឺម

- กำหนดให้เขียนโปรแกรมสำหรับการเชื่อมต่อระหว่าง GUI กับผู้ใช้งานในการบวก ลบ คูณ หารตัวเลขอย่างง่ายโดยมีรูปร่างของหน้า GUI ดังรูปตัวอย่าง



```

D:\2\cpe121\GUI\prepare for final\practice10_1.java
File Edit Tools Project Debugger Language Level Help
New Open Save Close Cut Copy Paste Undo Redo Find Compile Reset Run Test Javadoc Code Coverage

practice10_1.java
practice10_3.java
practice9_quiz.jav

1 import java.awt.*;
2 import javax.swing.*;
3 import java.awt.event.*;
4 public class practice10_1 extends JFrame{
5     public practice10_1(){
6         JLabel num1 = new JLabel("Number 1");
7         JTextField input1 = new JTextField(5);
8         JLabel num2 = new JLabel("Number 2");
9         JTextField input2 = new JTextField(5);
10        JLabel result = new JLabel("Result");
11        JTextField rs = new JTextField(5);
12
13        JPanel num = new JPanel(new FlowLayout());
14        num.add(num1);
15        num.add(input1);
16        num.add(num2);
17        num.add(input2);
18        num.add(result);
19        num.add(rs);
20
21        JButton ad = new JButton("ADD");
22        JButton sub = new JButton("Substract");
23        JButton mul = new JButton("Multiply");
24        JButton div = new JButton("Divide");
25
26        JPanel btn = new JPanel(new FlowLayout());
27        btn.add(ad);
28        btn.add(sub);
29        btn.add(mul);
30        btn.add(div);

```

— GUI

```

31
32        JPanel mainpanel = new JPanel(new GridLayout(2,1));
33        mainpanel.add(num);
34        mainpanel.add(btn);
35
36
37        add(mainpanel);
38
39        ad.addActionListener(new ActionListener(){
40            public void actionPerformed(ActionEvent e ){
41                Double numb1 = Double.parseDouble(input1.getText());
42                Double numb2 = Double.parseDouble(input2.getText());
43
44                Double res = numb1 + numb2 ;
45                rs.setText(Double.toString(res));
46            }
47        });
48
49        sub.addActionListener(new ActionListener(){
50            public void actionPerformed(ActionEvent e ){
51                Double numb1 = Double.parseDouble(input1.getText());
52                Double numb2 = Double.parseDouble(input2.getText());
53
54                Double res = numb1 - numb2 ;
55                rs.setText(Double.toString(res));
56            }
57        });

```

— event

```

58    }
59    };
60
61    mul.addActionListener(new ActionListener(){
62        public void actionPerformed(ActionEvent e ){
63            Double numb1 = Double.parseDouble(input1.getText());
64            Double numb2 = Double.parseDouble(input2.getText());
65
66            Double res = numb1 * numb2 ;
67            rs.setText(Double.toString(res));
68        }
69    });
70
71    div.addActionListener(new ActionListener(){
72        public void actionPerformed(ActionEvent e ){
73            Double numb1 = Double.parseDouble(input1.getText());
74            Double numb2 = Double.parseDouble(input2.getText());
75
76            Double res = numb1 / numb2 ;
77            rs.setText(Double.toString(res));
78        }
79    });
80
81    };
82
83

```

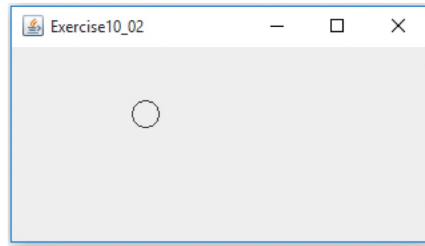
```

81    });
82
83    }
84
85    public static void main(String[] args){
86        practice10_1 f = new practice10_1();
87        f.setTitle("Exercise 10_01");
88        f.setSize(400,120);
89        f.setLocationRelativeTo(null);
90        f.setVisible(true);
91    }
92
93    }
94

```

} frame

2. กำหนดให้เขียนโปรแกรมสำหรับการเชื่อมต่อระหว่าง GUI กับผู้ใช้งานในการเลื่อนวงกลมในเฟรม โดยใช้ปุ่ม ลูกศรบนแป้นพิมพ์ในการบังคับทิศทางการเคลื่อนที่ไปซ้าย ขวา บน หรือล่าง ของวงกลมโดยมีรูปร่างของหน้า GUI ดังรูปตัวอย่าง



```
D:\2\cpe121\GUI\prepare for final\practice10_2.java
File Edit Tools Project Debugger Language Level Help
New Open Save Close Cut Copy Paste Undo Redo Find Compile Reset Run Test Javadoc Code Coverage

practice10_1.java
practice10_2.java
practice10_3.java
practice9_quiz.java

1 import java.awt.*;
2 import javax.swing.*;
3 import java.awt.event.*;
4
5 public class practice10_2 extends JFrame{
6     KeyboardPanel kb = new KeyboardPanel();
7     public practice10_2(){
8         add(kb);
9         kb.setFocusable(true);
10    }
11
12    public static void main(String[] args){
13        practice10_2 f = new practice10_2();
14        f.setTitle("Exercise 10_02");
15        f.setSize(400,400);
16        f.setLocationRelativeTo(null);
17        f.setVisible(true);
18    }
19
20    static class KeyboardPanel extends JPanel{
21        int x = 100,y = 100,diameter = 50;
22        public KeyboardPanel(){
23            addKeyListener(new KeyAdapter(){
24                public void keyPressed(KeyEvent e){
25                    switch (e.getKeyCode()){
26                        case KeyEvent.VK_DOWN: y+=10;
27                        break;
28                        case KeyEvent.VK_UP: y-=10;
29                        break;
30                        case KeyEvent.VK_LEFT: x-=10;
31                        break;
32                        case KeyEvent.VK_RIGHT: x+=10;
33                        break;
34                    }
35                    repaint();
36                }
37            });
38        }
39        protected void paintComponent(Graphics g){
40            super.paintComponent(g);
41            g.drawOval(x,y,diameter,diameter);
42        }
43    }
44 }
45
46 }
```

เพื่อรับค่าคำสั่งจาก Kb.

สร้าง fram

จุดเริ่มต้น

ขยับ

part graphic